

Assignment - 2

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1. Decision tree inconsistency in loan approval

Question

A bank deploys a decision tree model to approve loans. After deployment, regulators notice inconsistent decisions: two nearly identical applicants receive different outcomes.[page:3]

Tasks: a) Diagnose the issue in terms of decision tree instability and variance, b) Explain how small perturbations in training data affect split selection, c) Propose a redesign using Random Forest or Bagging, d) Discuss how ensemble averaging improves regulatory reliability.[page:3]

Answer

a) Decision tree instability and variance

A single decision tree is a high-variance learner, so small changes in the training data can produce a very different tree structure and therefore different predictions for nearly identical applicants. This inconsistency appears because greedy splitting decisions amplify minor sample differences across the full tree.

b) Effect of perturbations on split selection

At each node, the algorithm selects the split that maximizes impurity reduction such as information gain or Gini decrease. When two candidate splits have very similar impurity reductions, even a tiny perturbation in sample composition can change the chosen split, and that altered early split then changes all downstream branches.

For classification, a split typically maximizes

$$\Delta I = I(\text{parent}) - \frac{n_L}{n} I(\text{left}) - \frac{n_R}{n} I(\text{right})$$

If two candidate values of ΔI are close, tiny changes in data can reverse the ranking.

c) Redesign using Bagging or Random Forest

Bagging builds many trees on bootstrap samples and aggregates their predictions, reducing instability. Random Forest improves further by also sampling a random subset of features at each split, which makes trees less correlated and more robust than plain bagging.

d) Ensemble averaging and regulatory reliability

If the prediction from tree b is $T_b(x)$, the ensemble predictor is

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

Averaging reduces variance because random fluctuations from individual trees cancel out. This gives more stable decisions for similar applicants, which improves reliability, fairness, and defensibility in regulatory review.

2. Churn prediction with interpretability

Question

A telecom company wants a churn prediction model that is both highly accurate and easily interpretable for business managers.[page:3]

Tasks: a) Compare a deep decision tree vs pruned tree vs ensemble methods, b) Explain cost-complexity pruning in this business context, c) Discuss trade-off between interpretability and performance, d) Recommend a deployable solution with justification.[page:3]

Answer

a) Comparison of model types

Model	Advantages	Limitations	Business fit
Deep decision tree	Captures complex interactions	Overfits, unstable, hard to explain fully	Weak [page:3]
Pruned tree	Simpler, stable, readable rules	May lose some predictive power	Strong [page:3]
Ensemble methods	Usually highest accuracy	Harder to interpret	Moderate [page:3]

b) Cost-complexity pruning

Cost-complexity pruning minimizes

$$R_\alpha(T) = R(T) + \alpha|T|$$

where $R(T)$ is the tree error or impurity-based loss and $|T|$ is the number of terminal nodes. In business terms, the parameter α penalizes overly detailed rules that add complexity without enough practical value.

c) Interpretability versus performance

A deep tree or ensemble may produce slightly better accuracy, but business managers often need simple rules they can communicate and act on, such as low tenure plus repeated complaints implying churn risk. A pruned tree sacrifices some flexibility but provides transparent decision paths, making it easier to trust and deploy.

d) Recommended deployable solution

A pruned decision tree is the best production choice when interpretability is a hard requirement. A practical strategy is to benchmark an ensemble offline for maximum accuracy, then deploy a

pruned tree or rule-extracted surrogate that preserves most of the business value while remaining understandable.

3. Regression tree and abrupt demand predictions

Question

A retail company uses a regression tree for predicting product demand, but managers complain that predictions jump abruptly with small input changes.[page:3]

Tasks: a) Explain why regression trees produce piecewise constant outputs, b) Analyze limitations in capturing smooth trends, c) Compare with linear regression and boosting methods, d) Suggest a hybrid model and justify.[page:3]

Answer

a) Piecewise constant outputs

A regression tree partitions the feature space into regions $R_1, \dots, R_M$ and predicts a constant within each region:

$$\hat{f}(x) = \sum_{m=1}^M c_m \mathbf{1}(x \in R_m)$$

where c_m is usually the mean response in region R_m . Because the prediction is constant inside each region, crossing a split boundary causes an abrupt jump.

b) Limits in smooth trend modeling

Regression trees approximate functions by step-like segments, so they struggle when demand changes gradually with price, time, discount, or seasonality. This makes them poor at representing smooth response surfaces.

c) Comparison with alternatives

Linear regression produces continuous smooth predictions but cannot capture strong nonlinear interactions unless features are engineered. Boosting combines many small trees, creating a more flexible function that can approximate smooth nonlinear patterns much better than a single regression tree.

d) Recommended hybrid model

A good choice is gradient boosting with shallow trees or a model tree that fits a linear regression within each leaf. This approach keeps some local interpretability while reducing abrupt discontinuities and improving predictive smoothness.

4. Missing test results in hospital data

Question

In a hospital dataset, severely ill patients are more likely to have missing test results.[page:3]

Tasks: a) Identify the missing data mechanism, b) Explain how this biases decision tree splits, c) Compare surrogate splits vs imputation techniques, d) Recommend a robust modeling pipeline.
[page:3]

Answer

a) Missing data mechanism

This is an MNAR situation, meaning Missing Not At Random, because the probability of missingness depends on illness severity or on the unobserved variable itself. The missingness therefore contains signal related to the outcome.

b) Bias in tree splits

If missing values are ignored or poorly imputed, the split criterion becomes biased because the missingness pattern is not random. The tree may incorrectly associate the absence of a measurement with a lower-risk or higher-risk subgroup depending on how the data are handled.

c) Surrogate splits versus imputation

Surrogate splits replace the main split variable with another correlated variable when the original feature is missing, so they preserve the tree structure but depend on strong substitutes being available. Imputation fills missing values before modeling, which can be simple and efficient, but naive methods like mean imputation may wash out clinically important uncertainty in MNAR data.

d) Robust pipeline

A robust pipeline should first analyze missingness patterns, then create missingness indicators, apply clinically informed or multiple imputation, and train models with careful validation. Using an ensemble tree model together with missing-value flags and sensitivity analysis is usually safer than relying on a single tree.

5. Fraud detection with 0.2% fraud cases

Question

A digital payment system detects fraud with only 0.2% positive cases, yet the deployed model shows 99.8% accuracy and still misses fraud badly.[page:3]

Tasks: a) Explain why accuracy is misleading, b) Analyze model performance using ROC and Precision-Recall curves, c) Discuss threshold tuning under cost-sensitive learning, d) Recommend evaluation and model improvement strategy.[page:3]

Answer

a) Why accuracy is misleading

If only 0.2% of cases are fraud, a dummy classifier that predicts every transaction as non-fraud already achieves 99.8% accuracy. This means accuracy says almost nothing about actual fraud-catching ability in an extreme class imbalance setting.

b) ROC and Precision-Recall analysis

The ROC curve plots True Positive Rate against False Positive Rate across thresholds and is useful for overall ranking quality. The Precision-Recall curve is more informative here because it focuses on how well the model retrieves the rare fraud class without producing too many false alarms.

c) Threshold tuning under cost-sensitive learning

The threshold should not stay fixed at 0.5 when false negatives are much more expensive than false positives. Instead, the decision rule should be chosen to minimize expected cost, which usually means lowering the threshold to increase recall for fraud.

d) Recommended strategy

Use recall, precision, F1-score, PR-AUC, ROC-AUC, and a confusion matrix instead of relying on accuracy alone. Improve the model with class weighting, resampling, threshold optimization, calibrated probabilities, and fraud-specific feature engineering tied to transaction velocity, device behavior, and anomalous patterns.

6. Bagging with highly correlated trees

Question

A financial firm uses bagging to stabilize predictions, but results remain highly correlated across trees.[page:3]

Tasks: a) Explain why bagging fails with correlated features, b) Analyze how Random Forest reduces correlation via feature sampling, c) Discuss impact on bias and variance, d) Recommend improvements for better generalization.[page:3]

Answer

a) Why bagging may fail

Bagging only helps strongly when the individual trees are diverse. If the dataset contains dominant correlated features, many bootstrap samples still choose the same top predictors near the root, so the resulting trees remain similar and the ensemble correlation stays high.

b) How Random Forest reduces correlation

Random Forest samples only a subset of features at each split, so even strong predictors are sometimes excluded from consideration. This forces different trees to explore different structures and lowers inter-tree correlation, which improves the gain from averaging.

c) Impact on bias and variance

Feature subsampling may slightly increase bias because some trees cannot always use the

strongest split. However, the variance usually decreases much more, so the overall test error often improves.

d) Recommended improvements

Switch from bagging to Random Forest, tune the number of candidate features per split, control tree depth, and reduce redundant predictors through feature engineering. Extra Trees, decorrelated embeddings, or grouped feature sampling can further improve generalization when correlation is severe.

7. Gradient Boosting with noisy click labels

Question

An online advertising system uses Gradient Boosting to predict ad clicks, but many labels are noisy due to accidental clicks.[page:3]

Tasks: a) Explain how boosting emphasizes misclassified points, b) Analyze why it overfits noisy data, c) Compare with bagging robustness, d) Suggest regularization strategies.[page:3]

Answer

a) Why boosting emphasizes errors

Boosting builds models sequentially, with each new learner focusing on observations that previous learners predicted poorly. In gradient boosting, the next tree fits the negative gradient of the loss, so large residual-error points receive more corrective attention.

b) Why noisy labels cause overfitting

If some clicks are accidental, those labels may not represent true user intent. Boosting repeatedly tries to fit these hard mislabeled cases, so later rounds may learn noise rather than genuine click behavior.

c) Comparison with bagging

Bagging is usually more robust to noisy labels because it trains trees independently and averages them instead of aggressively chasing residual mistakes. Boosting can outperform bagging on clean data, but it is typically more sensitive to label noise.

d) Regularization strategies

Use a smaller learning rate, shallow trees, early stopping, row subsampling, column subsampling, and robust loss tuning. These controls reduce the influence of noisy observations and improve test performance.

8. Stacking and test-set performance drop

Question

A company combines Decision Trees, SVM, and Logistic Regression using stacking, but training accuracy is high while test performance drops significantly.[page:3]

Tasks: a) Diagnose data leakage in stacking, b) Explain importance of cross-validation in meta-model training, c) Compare stacking vs bagging vs boosting, d) Propose a corrected architecture. [page:3]

Answer

a) Data leakage diagnosis

A frequent stacking mistake is training the meta-model on base-model predictions generated from the same data that were used to train the base models. Those predictions are overly optimistic, so the meta-model learns from leaked information and fails on unseen data.

b) Importance of cross-validation

Proper stacking uses out-of-fold predictions from cross-validation. For each fold, the base learners are trained on the remaining folds and then predict the held-out fold, so the meta-model only sees predictions that simulate true unseen performance.

c) Stacking vs bagging vs boosting

Method	Core idea	Strength	Main risk
Bagging	Parallel bootstrap models averaged together	Reduces variance	Less gain when models are correlated [page:3]
Boosting	Sequentially focuses on previous errors	High predictive power	Sensitive to noise and overfitting [page:3]
Stacking	Combines different model families with a meta-learner	Exploits complementary strengths	Leakage if meta-training is done incorrectly [page:3]

d) Corrected architecture

Use K-fold stacking to generate out-of-fold predictions from Decision Tree, SVM, and Logistic Regression, then train a simple meta-model such as logistic regression on those predictions. After that, refit the base models on the full training set and feed their test predictions into the already trained meta-model.

9. Soft-margin SVM for tumor classification

Question

A dataset for tumor classification contains overlapping benign and malignant cases.[page:3]

Tasks: a) Formulate the soft-margin SVM optimization problem, b) Explain the role of slack variables and penalty parameter, c) Interpret hinge loss in medical decision-making, d) Discuss the trade-off between sensitivity and specificity.[page:3]

Answer

a) Soft-margin SVM optimization

For labeled samples (x_i, y_i) with $y_i \in \{-1, +1\}$, soft-margin SVM solves

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

subject to

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

This balances a large margin with tolerance for overlap and classification violations.

b) Slack variables and penalty parameter

The slack variables ξ_i measure margin violations or outright misclassifications. The parameter C controls how strongly those violations are penalized, with large C favoring strict separation and small C allowing a softer, more regularized boundary.

c) Hinge loss in medical decision-making

The hinge loss is

$$\ell_i = \max(0, 1 - y_i f(x_i))$$

It penalizes not only wrong classifications but also correct predictions made with weak margin. In medicine, this is useful because a tumor case predicted as malignant with low confidence may still require caution and further testing.

d) Sensitivity and specificity trade-off

Sensitivity measures how many malignant cases are correctly detected, while specificity measures how many benign cases are correctly rejected. In tumor screening, sensitivity is often prioritized because missing a malignant case is usually more harmful than sending some benign cases for additional evaluation.

10. Naive Bayes with co-occurring words

Question

A spam detection system uses Naive Bayes, but certain words frequently co-occur, such as “offer” and “free.”[page:3]

Tasks: a) Explain violation of conditional independence assumption, b) Analyze impact on posterior probability estimation, c) Compare with Bayesian Networks, d) Suggest improvements.
[page:3]

Answer

a) Violation of conditional independence

Naive Bayes assumes

$$P(x_1, \dots, x_d | y) = \prod_{j=1}^d P(x_j | y)$$

which means the features are conditionally independent given the class. When words like “offer” and “free” co-occur frequently, that assumption is violated because one word gives information about the presence of the other even after conditioning on spam status.

b) Effect on posterior estimates

Naive Bayes may double-count evidence by multiplying correlated likelihood terms as though they were independent. This often leads to overconfident posterior probabilities and weaker probability calibration.

c) Comparison with Bayesian Networks

Bayesian Networks are more expressive because they explicitly model dependencies among variables using a directed graph. They can capture word co-occurrence structure better than Naive Bayes, but they are also more complex to learn and infer from.

d) Suggested improvements

Use bigrams or n-grams to model common word pairs directly, remove redundant features, apply Complement Naive Bayes, or switch to logistic regression, linear SVM, or a Bayesian Network when dependencies between words are important. These methods handle correlated text features more effectively than plain Naive Bayes.